A MERGE-Based Derivation of Tagalog/Filipino Basic Sentence Structure: A Preliminary Sketch

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Abstract

The Strong Minimalist Thesis (SMT) is understood as a requirement that the structures of I-language are generated by the simplest operations, and that the faculty of language is an optimal solution to certain language-specific conditions (LSCs) (Chomsky et al., 2023). Using SMT as a guiding principle, Chomsky and others proposed a new formulation of MERGE that applies on the workspace (WS)—a set consisting of inscriptions of lexical items available for computation at a given derivational stage—instead of directly manipulating syntactic objects This formulation of MERGE is constrained by general principles that govern the operation of a computational mechanism known as third factor constraints. Following MERGE and SMT, the present study offers a preliminary analysis of Tagalog/Filipino sentences, particularly the transitive and intransitive constructions, wh-question formation, and equative sentences.

Keywords: Strong minimalist thesis, MERGE, third factor constraints, Tagalog, equative sentences

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Tagalog is one of the Austronesian languages widely spoken in the capital region of the Philippines. It is the basis of Filipino, the Philippine National Language, henceforth Tagalog/Filipino. Tagalog/Filipino is a verb-first language known for its flexible word order. There have been many proposed analyses of the basic sentence structure of this language within the tradition of generative grammar such as those found in Aldridge (2006), and Ceña and Nolasco (2011) among others. Although the derivations proposed in these works may still be adopted for syntactic description, they are not aligned with the current conception of generative grammar i.e., new formulation of MERGE and the Strong Minimalist Thesis (SMT), which undermines the goal of grammar of being "concerned with explanation, not mere description" (Chomsky et al., 2023, p. 09). Using the new formulation of MERGE, I propose a preliminary sketch of how Tagalog/Filipino sentences can be analyzed using this formulation. This preliminary analysis draws from Aldridge's (2006) analysis of Tagalog/Filipino sentence structures, and reconceptualizes them in the terms of MERGE and SMT. Additionally, I briefly discuss equative sentences found in Ceña and Nolasco (2011) and argue that their conclusion that equative sentences are a result of movement is correct. I derive their conclusion from the distinct application of MERGE and the duality of semantics—a language specific condition that states that external merge (EM) and only EM can create a theta position. This shows that the new formulation of MERGE could provide a principled choice between two competing syntactic analyses of the same phenomenon. The present study contributes to the goal of using less commonly studied languages in illustrating, sharpening, and refining theoretical concepts through their application in the analyses of these languages.

Brief Review of the Historical Development of MERGE

In 1957, Chomsky formulated generative grammar by pointing out the weaknesses of the grammatical model for syntax called as Finite State Grammar. He showed that this type of grammar is incapable of capturing the recursive property of language. He also added that finite state grammar also lacks tools in capturing long distance dependency or relation in a sentence—an argument echoed by Pinker (1984). Chomsky then proposed his own grammatical model for syntax known as Generative Grammar and formulated the Phrase Structure Rules or PS rules. These rules can be written in a rewrite format, bracketing or tree diagram also known as phrase markers. These rules specify the immediate constituents of the sentence as phrases and words. Thus, a sentential analysis that uses PS has the following representation.

(1) Clause→ Phrase→ Word
Following this format, a phrase structure rule for sentence is:
(2) S→NP ⁀VP
This rule can be applied to phrases. Thus, a rule for nominal phrase can be written as:
(3) NP→Det ⁀N
And the rules for verb phrase and prepositional phrase are written in (4) and (5) respectively as
(4) VP→ V ⁀NP
(5) PP → P ⁀ NP

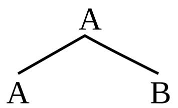
The theory of Phrase Structure Grammar shows not only the hierarchical order of words in a sentence, which contributes to its whole meaning, it also shows relation between the words in a sentence. Structural relations such as Dominance, Sisterhood, Precedence, C-command and Government, under which many grammatical relations are assigned, can be better realized with the use of the Phrase Structure Grammar. Phrase Structure Grammar, however, eventually faced several problems which led to its revisions. Subsequent revisions were anchored on the need for grammatical model to account for the ease and rapidity of language acquisition encapsulated in the concept of explanatory adequacy. Under the lens of explanatory adequacy, the phrase structure rules were deemed too complex to account for language acquisition. This, in turn, led to the formulation of X-bar theory which considerably reduced the rules of phrase structure grammar. The X-bar theory, which replaces the PS rules as combinatorial component of grammar, became one of the two central operations in the 1981's framework Government and Binding Theory (henceforth GB).

Another important syntactic concept introduced in Generative Grammar along with phrase structure rules is transformations. Originally introduced by Zellig Harris in 1950s, Chomsky (1957) adopted this theoretical device to generate structures that Phrase Structure Grammar fails to describe. This operation takes the form of Σ to Σn, Σ being the structure generated by Phrase Structure and Σn being the output of the application of transformation (Chomsky, 1957). Transformation adds, deletes, and move elements within the phrase markers. Chomsky identified many types of transformations such as the following: DP raising, Passivization, Wh-movement, Head Movement and many more. Wh-question formation in English for example can be described by a series of transformations such Wh-movement, Head Movement, and Do-support. Transformations were also reduced to the basic operation move α in the GB framework. X-bar theory and move α were both considered as components of Universal Grammar—the language module in the brain.

Both X-bar theory and move α were further subject to modifications as generative grammar shifted its focus from descriptive and explanatory adequacy to explanatory adequacy (learnability in the current framework) and evolvability—the question of how language ability emerged in humans. This shift began in Chomsky's (1995) Minimalist Program and developed in his subsequent works. This culminated to the proposal that the combinatorial operation of language can be accounted for by a single operation known as Merge. Merge can simply be defined as an operation that combine two elements in an unordered set as shown in (6).

(6) a. Merge (A, B) = {A, B}

b.



The syntactic objects A and B are selected as the input for Merge in (6a). The output of Merge is an unordered set {A, B}. (6b) shows the tree diagram representation of the output of Merge. Merge was formulated in the Bare Phrase Structure Grammar (BSP) which eliminated bar levels introduced in the X-bar theory (Alexiadou & Lohndal, 2021). Subsequent works also aimed at further minimizing the operations in syntax in terms of their number or their complexity which resulted in approaches where labeling and other concepts were derived elsewhere other than from Universal Grammar. In keeping with this aim, Chomsky (2006) claimed that transformation or move α can be formulated as an application of Merge. This application of Merge selects a component of SO and merges it to the SO itself. This type of Merge is shown in (7)

(7) a. Merge (A, B) = {A, B} where a copy of A is a member of B

The reduction of move α to Merge further minimizes the syntactic system of language. Merge alone can account for both combinatorial and transformational aspects of language. The application of Merge that combines two SOs is known as External Merge (henceforth EM) while the one that applies to move a component of SO and merge it to itself is known as Internal Merge (henceforth IM). Aside from these two applications of Merge, several extensions of Merge have been proposed in the literature. One such example, which resulted from Chomsky (2000, 2001, 2008), is pair merge. It combines two elements to form an ordered set. Pair merge accounts for phrasal adjunctions such as adverbial and adjectival constructions and head-to-head incorporation. Other iterations of Merge are late merge (Lebeaux, 1988), sideward merge (Nunes, 2001), and parallel merge (Citko, 2000, 2005; Citko & Yuksek, 2020).

The reduction of language to its basic operation is not only motivated by the strong minimalist thesis and the explanatory adequacy, but also by the biolinguistic desideratum of identifying the phenotype that underlies language ability in humans allowing scientific inquiry into its evolution (Berwick & Chomsky, 2016). The reduction to Merge also relegated the complexities of syntactic structures to other factors that cannot be derived directly from Merge. These factors, along with Merge, are one of the components of Chomsky's three factor approach to the growth of language in humans.

The Three-factor Design of Human Language

Chomsky (2005, 2007, 2008) proposed the three-factor design approach to language that postulates three factors that underlie the structure and development of human language faculty: the biological or genetic endowment, external data or experience, and the general laws of computation also known as the third-factor constraint. The first factor refers to the operation attributable to Universal Grammar. This operation is Merge—a combinatorial operation that yields {X, Y} from X and Y (Berwick & Chomsky, 2016). The second factor, on the other hand, refers to the language input that children are exposed to during the language learning. The last of the three factors is the principles of structural architecture and development constraints. These principles, also known as 3rd factor principles, are not attributable to language specific conditions or Merge but instead can be derived from the laws of nature.

3rd Factor Compliant MERGE

In line with the three-factor design and the SMT, Chomsky et al. (2017), Chomsky (2019), Chomsky (cited in Bencheikh, 2019), Epstein et al. (2022) and Chomsky et al. (2023) proposed a new formulation of Merge. In this new formulation, Merge, written as MERGE, is described as an operation that applies on a Workspace—a set consisting of inscriptions of lexical items available for computation at a given derivational stage.

(8) MERGE (Epstein et al., 2022; Chomsky et al., 2023)

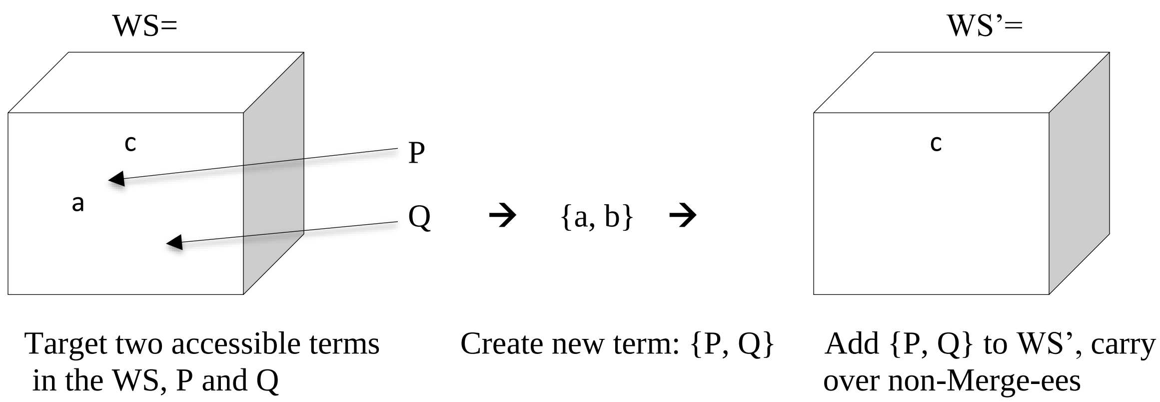
MERGE (WS)→WS'

MERGE (P, Q, WS) = WS' = [{P, Q}, …]

where the ellipsis … stands for elements in the WS other than P and Q.

The figure in (9) illustrates how MERGE operates on the elements a and b.

(9) *Illustration of WS, the Application of MERGE, and the Resultant WS' Reproduced from Epstein et al. (2022, p. 121)*



In (9), Workspace (WS) is composed of inscriptions of lexical items from the lexicon available to syntactic computation (Epstein et al., 2023) while MERGE is an operation that manipulates the Workspace (WS) instead of the syntactic objects directly. When MERGE applies to a and b which are considered as P and Q, it results in a Workspace (WS') containing a set, {P, Q}, and other elements still available for computation. To further illustrate how MERGE works, take for example the derivation of the sentence, *The man reads a book.*

(10) a. WS= [a, book]
b. MERGE (a, book, WS) = WS' = [{a, book}, …]
c. WS = [R, {a, book}, …] where R= read
d. MERGE (R, {a, book}, WS) = WS' = [{R, {a, book}}, …]
e. WS = [v*, {R, {a, book}}]
f. MERGE (v*, {R, {a, book}}, WS) = WS' [{v*, {R, {a, book}}}, …]
g. WS = [{the, man}, {v*, {R, {a, book}}}]
h. MERGE ({the, man}, {v*, {R, {a, book}}}, WS) = WS' = [{({the, man}, {v*, {R, {a, book}}}, …]

i. WS = [T, v*P]
j. MERGE (T, v*P, WS) =WS' = [{T, vP}, …]
k. WS' = [{T, v*P}]
l. MERGE (EA, {T, v*P}, WS) = WS' = [{EA, {T, v*P}}, ⋯] where EA= {the, man} ∈ v*P

In this derivation, the WS initially contains members, *a* and *book* in (10a). MERGE is then applied to WS to form the set {a, book} in (10b). The root R is then added to WS from the lexicon in (10c) and merged in (10d). This is followed by the addition of v* to WS in (10e) and its subsequent merging in (10f). The external argument, EA = {the, man}, is then added to the WS in (10g) and merged in (10h). The derivation proceeds with the addition T head in WS as shown in (10i) and its merging in (10j) with the resulting WS' in (10k). The derivation ends when the EA is internal merged with the rest of the structure in (10l).

Throughout the derivation, there is only one postulated structure building operation MERGE. Epstein as cited in DMC KeioUniversity (2023) asserted that this MERGE as formulated is an operation in a vacuum and that it needs to be constrained by 3rd factor principles. These 3rd factors principles are general principles that govern the operation of computational mechanism (Chomsky et al, 2023)

3rd Factor Principles

Epstein et al. (2022) and Chomsky et al. (2023) discussed several 3rd factor principles that govern the application of MERGE. I list them down in (11) and discuss each in following paragraphs.

(11) a. Binarity
b. Minimal Search
c. Preservation
d. Uniform Interpretation of Identical Inscription
e. Minimal Yield
f. No Tampering Condition or NTC
g. Phase Impenetrability Condition or PIC

(11a) is a principle that limits the application of MERGE to only two inscriptions or SOs. The requirement of MERGE to apply on only two inscriptions or SOs is the simplest form of MERGE. Chomsky et al (2023) argue that binary MERGE along with other third factor constraints, is based on the principle of Simplicity-by-least-effort, and that it is more economical than non-binary operation i.e., "less computation is simpler than more" (p. 17).

The 3rd factor in (11b) requires MERGE to locate its target inscription or SOs in a stepwise fashion. MERGE first locates members of the WS before the terms of the members where term is defined as:

(12) X is a term of Y iff X is either (i) a member of Y or (ii) a member of a term of Y (Chomsky et al., 2023, p. 19).

Based on this definition, if the WS contains [a, {b, c}], a and {b, c} are both members and terms of the WS but b and c are just terms. Minimal search dictates that a and {b, c} are available to the first Search step as these two inscriptions are members of the WS not terms. Accessing the terms b and c in {b, c} in the first Search step requires more computation thus violates the principle of Simplicity-by-least-effort. Terms are accessed in second search step in the application of IM.

Preservation in (11c) ensures that the interpretation of an inscription remains constant throughout the computation. Unlike binarity and minimal search, Preservation is not Markovian which means that it can look into previous stages of derivation to determine whether the meaning of inscription has changed. To illustrate this 3rd factor principle, take for example the following WS in (13).

(13) a. WS = [{was, {eaten, {the, apple}}}] *Taken from Chomsky et al. (2023, p. 22)*

b. WS' = [{{the, apple}, {was, {eaten, {the, apple}}}}]

Since preservation can see both the WS and WS', it can determine that the inscription {the, apple} has the same interpretations in both input WS and output WS'.

Uniform Interpretation of Identical Inscription or UIII in (11d) may appear to be a duplication of (11c) but according to Chomsky et al. (2023), the two should not be confused. Preservation ensures that the interpretation of an inscription does not change in the course of computation while UIII ensures that identical inscriptions receive the same interpretation. The UIII plays an important role in the operation Form Copy (FC). FC assigns copy relation to two structural identical inscriptions. Take for example the following WSs in (13).

(14) a. WS = [{{many, people}, {praised, {many, people}}}]

b. WS = [{{many, people}, {were, {praised, {many, people}}}}]

In (14a), EM is applied to combine the SOs, {many, people} and {praised, {many, people}} while in (14b), IM applies taking {many, people} from {were, {praised, {many, people}}} and combining two SOs together. Once a certain point of the derivation is reached, e.g., a phase, FC can apply to the WS.

(15) Form Copy (Chomsky et al., 2023, p. 24)
Where X, Y are structurally identical, FC (X, Y) interprets X, Y as copies, that is, the inscriptions are interpreted in exactly the same way.

Once the copy relation is assigned to the structurally identical inscriptions, the UIII ensures the two receives the same interpretation. The WS in (14a) is not governed by UIII as FC cannot assign copy relation to the SOs {many, people} in the subject position and {many, people} in the complement position. When FC fails to apply on two similar inscriptions, they are considered as repetitions which allows for different interpretations.

The Minimal Yield or MY in (11e) is an instantiation of the general principle Restrict Computational Resources or RCR. The RCR stems from Fong's (2021) idea that the brain is not a computational powerhouse. MERGE, being a component of the brain, must be governed by RCR. Epstein et al. (2022) directly applies RCR to MERGE and defines it as restriction on the cardinality of WS i.e., WS cannot expand its cardinality; the number of members in a set cannot increase. Chomsky et al. (2023) derive this restriction from MY which limits the number of new accessible terms in WS. To see how MY applies, take for example the WS in (16).

(16) a. WS = [a, b, c]

b. MERGE (b, c, WS) P= b; Q= c *Taken from Chomsky et al., (2023, p. 29)*

c. WS' = [a, b, c, {b, c}]

The derivation in (16) is a violation of MY as it increases the number of new accessible terms in the WS. The inscriptions {b, c}, b and c are available for further computation. The b and c in (16a) are merged in (16b) thus the b and c are new inscriptions in (16c) which increases the accessible terms. Epstein et al.'s (2022) definition of RCR is captured by MY as increasing the accessible terms also increases the cardinality of WS i.e., the input WS has three members or terms while the output WS' has four. This instantiation of RCR as MY also captures determinacy which requires accessible terms to appear only once in the WS. It precludes multiple copies of a term in WS so that the subsequent operations can apply in a determinate fashion (Epstein et al., 2022)

The 3rd factor constraint in (11f), the No Tampering Condition or NTC, states that no change shall be imposed on SOs (Chomsky, 2008). One of the consequences of NTC is the elimination of linear order from the narrow syntax which has been relegated to the externalization. The linear order of SOs was implicit in the early formulation of phrase structure rules, and parameterized in X-bar theory. From this, NTC also rules out other extensions of Merge particularly pair merge which creates an ordered set from two elements combined.

The last of the 3rd factor constraint is the Phase Impenetrability Condition or PIC given in (17).

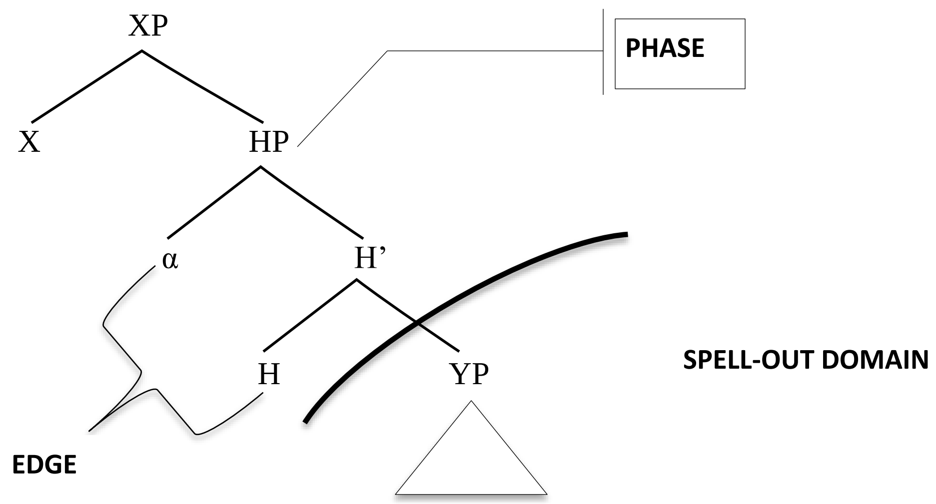
(17) Phase Impenetrability Condition (Chomsky, 2000)

In a phase α with the head H, the domain H is not accessible to operations outside α, only H and its edge are accessible to such operations.

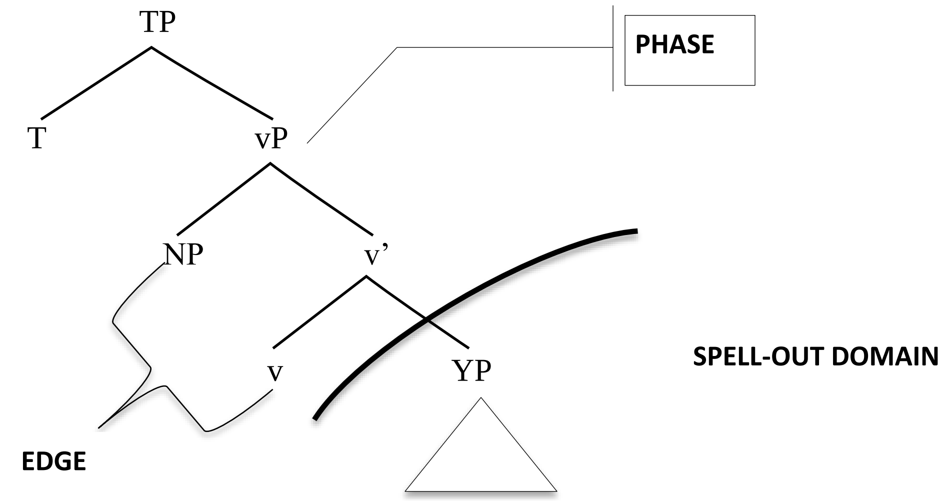
In the Minimalist Program, H is either v* or C. PIC prevents syntactic operations e.g., movement to apply on the domain or the complement of v* or C. Citko (2014) illustrates PIC in the following structures in (18a) and (18b)

(18) *Tree Diagrams Illustrating Phase and the Constraint on the Phase's Domain Reproduced from Citko (2014, p. 32)*

a.



b.



The domain of a phase is sent to the sensorimotor (externalization either spoken or signed) and conceptual-intensional interfaces (meaning or interpretation). Any syntactic operation is barred from applying to this domain.

Language Specific Conditions

In addition to MERGE and the 3rd factor principles, Chomsky et al. (2023) also proposed language specific conditions (LSC),

(19) a. Theta Theory
b. Duality of Semantics

(19a) refers to the thematic relations of the verb and its arguments while the (19b) states that external MERGE is the only operation that creates theta position.

The concepts of MERGE, 3rd factor constraints, and language specific conditions constitute the three-factor approach to language design of Chomsky (2005, 2007, 2008). These components work together to give rise to the computational complexity that the human language exhibits. The development of these concepts was motivated by empirical, theoretical, and epistemological consideration.

Method

This paper employs analytic and armchair method. Analytic method is defined by Chametzky (1999) as, "concerned with investigating the (phenomenon) domain in question. It deploys and tests concepts and architecture developed in theoretical work, allowing for both understanding of the domain and sharpening of the concepts." (p. xviii). This paper aims to apply the current formulation of MERGE and the strong minimalist thesis in offering a preliminary analysis of the sentence structures of Tagalog/Filipino The example sentences were contrived and their grammatical status were based on the intuition of the author. This method of imagining sentences for the linguist's judgment is known as the armchair method. Other examples and previous analyses of Tagalog/Filipino sentences were also adopted in the proposed derivations particularly from Ramos (1971), Aldridge (2006), and Ceña and Nolasco (2011).

Analysis

One-Place Predicate

The basic sentence structure of Tagalog/Filipino is VSO/VOS. It puts the verb initially followed by its arguments, either by the subject or object. One-place predicate sentence in Tagalog/Filipino is illustrated in (20).

(20) a. T <um> akbo ang bata.
<EA. foc> PERFC.INTR-run ABS child *Taken from Ramos (1971, p. 77)*
The man ran.

In MERGE type derivation, the WS first contains two inscriptions, v and R=*takbo.* MERGE then applies and form the set {v, R}

(21) a. WS= [v, R] where R= takbo
b. MERGE (v, R,, WS) = WS' = [{v, R}, …]

The derivation proceeds with the addition of *bata* to the WS.

c. WS = [bata, {v, R}]
d. MERGE (bata, {v, R}, WS) = WS' = [{bata, {v, R}}, …]

Following Aldridge (2006), the inscription *lalake* is assigned absolutive case by T.

e. MERGE (T, {bata, {v, R}}, WS) =WS' = [{T, {bata, {v, R}}, …]

In other types of intransitive constructions where the sole argument is a patient or an experience, the derivation begins with the WS containing the Root and the patient or experiencer argument.

(22) D-um-ating ang babae.
<IA. Foc> PERFE. INTR-arrive ABS woman *Taken from Aldridge (2006)*
The woman arrived.

(23) a. WS= [R, babae] where R= dating
b. MERGE (R, babae, WS) = WS' = [{R, babae}, …]
c. WS= [v, {R, babae}]
d. MERGE (v, {R, babae}, WS) = WS' = [{v, {R, babae}},…]
e. MERGE (babae, {v, {R, babae}}, WS) = WS' = [{babae, {v, {R, babae}}},…]
f. WS= [T, {babae, {v, {R, babae}}}]
g. MERGE (T, {babae, {v, {R, babae}}}, WS) = WS' = [{T, {babae, {v, {R, babae}}}},…]

As could be noticed in the preceding derivations, the classical analysis of verb movement where V (R in the present analysis) raises to v and then V+v raises to T is not integrated. This type of movement is only possible via pair-merge—an operation used by Chomsky and others to capture such movement. However, like other extensions of Merge, pair-merge is illicit under the current framework as it violates the 3^{rd} factor constraint, NTC. Pair merge creates an ordered set. As an alternative to this movement analysis, I assume that the string of heads T, v, and R is processed by a post-syntactic morphological operation at the externalization interface resulting in a morphological form reflecting the features of T and v such focus, aspect ang transitivity. I leave the details of this alternative to future research.

Two-place Predicate

In Tagalog, there two types of two-place predicate structure, Transitive Constructions and Intransitive Construction. In both of these constructions, sentences must have two arguments to be grammatical.

(24) a. Binasa ng mag-aaral ang teksto.
<IA. foc> PERFC.TRV-read ERG student ABS text
The student read the text.
b. *Binasa ang teksto.
<IA. foc> PERFC.TRV-read ABS text
*Read the text.
c. *Binasa ng mag-aaral
<IA. foc> PERFC.TRV-read ERG student
*The student read

In addition, each of the arguments in both of these constructions is marked by either ANG or NG markers.

(25) a. Bumili ang guro ng tela.
<EA. foc> PERFC.INS-buy ABS teacher ERG cloth
The teacher bought the cloth.
b. Binili ng guro ang tela.
<IA. foc> PERFC.TV-buy ERG teacher ABS cloth
The teacher bought the cloth

These markers, which are sometimes classified as prepositions, indicate which of the arguments receives focus. This marking pattern on arguments is related to the verb morphology. For instance, when the verb bears the infix *-um* and *mag* the external argument (EA) is preceded by ANG while the internal argument (IA) is preceded by NG as shown in (25a). In contrast, when the verb bears the *-in* infix, the IA is the argument preceded by ANG while the EA is preceded by NG (25b). The relationship between these argument marking patterns and verb morphology is one of the long-standing problems in Tagalog/Filipino syntax. The argument marking patterns also influences the availability of these arguments to be subject to further syntactic operations i.e., the argument that bears the ANG marker is the one that can be extracted in *wh*-questions and equative sentence formation.

(26) *wh*-questions in Tagalog
a. Sino ang bumili ~~ang guro~~ ng tela?
Who ANG <EA. foc> PERFC.INS-buy ERG cloth?
Who bought the cloth?
b. Ano ang binili ~~ang tela~~ ng guro?
What ANG <IA. foc> PERFC.TV-buy ERG teacher

(27) Equative sentence constructions
a. Ang guro ang bumili ~~ang guro~~ ng tela.
ABS teacher ANG <EA. foc> PERFC.INS-buy ERG cloth.
It is the teacher who bought the cloth.
b. Ang tela ang binili ~~ang tela~~ ng guro.
ABS cloth ANG <EA. foc> PERFC.INS-buy ERG teacher.
It is the cloth that the teacher bought.

Both Aldridge (2006) and Ceña and Nolasco (2011) capture this relationship between the marking pattern on arguments and verb morphology as an agreement formed between T or v and the arguments; *-um* forms an agreement with the EA while *-in* forms agreement with IA. Aldridge (2006) also added that the ANG-marked argument in the case of transitive verb i.e., verb with *-in* infix, raises to the edge v*P.

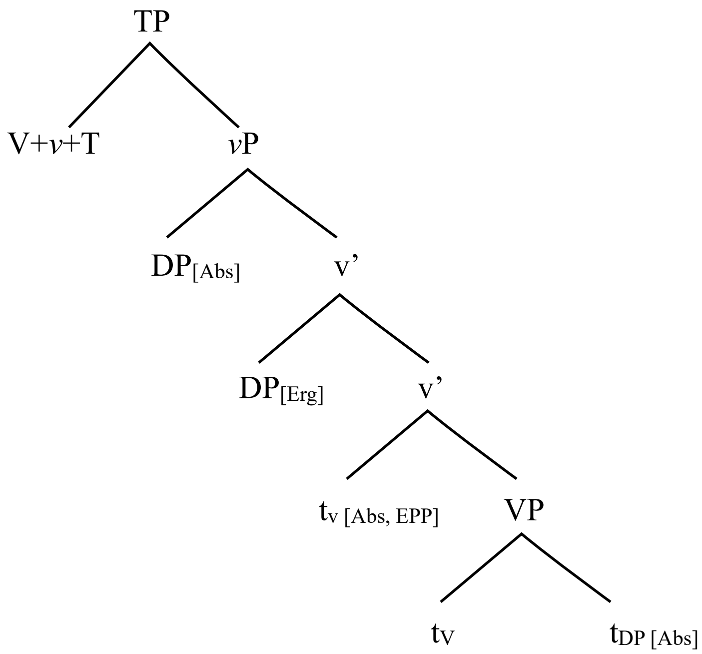
Using the current formulation of MERGE, a transitive construction like in (28) is formed using the following derivation in (29).

(28) a. Binasa ng mag-aaral ang teksto.
<IA. foc> PERFC.TRV-read ERG student ABS text
The student read the text.

(29) a. WS= [R, teksto] where R=basa
b. MERGE (R, teksto, WS) = WS'= [{R, teksto}, …]
c. WS= [v*, {R, teksto}]
d. MERGE (v*, {R, teksto}, WS) = WS'= [{v*, {R, teksto}}, …]
e. WS = [EA, {v*, {R, teksto}}]
f. MERGE (EA, {v*, {R, teksto}}, WS)= WS' = [{EA, {v*, {R, teksto}}},…]
g. MERGE (teksto, {EA, {v*, {R, teksto}}}, WS) =WS' = [{teksto, {EA, {v*, {R, teksto}}}}
h. WS= [T, {teksto, {EA, {v*, {R, teksto}}}}]
i. MERGE (T, {teksto, {EA, {v*, {R, teksto}}}}, WS) =WS' = [{T, {teksto, {EA, {v*, {R, teksto}}}}}

The derivation starts with the Root and *teksto* inside the WS in (29a). MERGE then applies in (29b) resulting in {R, teksto}. The v* is then added in the WS in (29c) and merged to with {R, teksto} in (29d). After this, EA is added to WS and merged with {v*, {R, teksto}} in (29e) and (29f). *teksto* is then internal merged in (29g) which accounts for the movement analysis of Aldridge (2006) in which the IA moves to outer edge of v*P as shown in the following tree diagram.

(30) *Tree Diagram Illustrating Transitive Construction Adapted from Aldridge (2006)*



where $DP_{[abs]}$ is the internal argument NP=teskto in the present analysis.

This IM of internal argument from the complement position of the root, which is labeled as V in Aldridge (2006), conforms to the current formulation of MERGE and the 3rd factors principles. Minimal yield, for instance, is followed since the lower copy of the IA is rendered inaccessible by PIC once the phase v*P (*v*P in Aldridge's analysis) is complete. Additionally, the lower copy is not pronounced as FC assigns copy relation to the two NPs.

(31) a. FC (teksto, teksto)

b. WS= {T, {teksto, {EA, {v*, {R, ~~teksto~~}}}}}

In the case of Intransitive construction marked by *-um* or *mag-*, ANG is placed before the Actor of the sentence as shown here in (31).

(32) a.

Bumasa	ang	mag-aaral	ng	teksto.
<EA. Foc> PERFC.INTR-read	ABS	student	OBL	text.

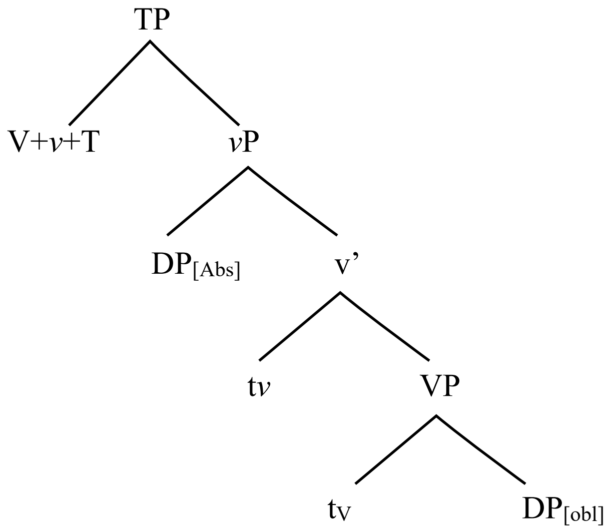
The student read the text.

The derivation for this construction using the current formulation of MERGE is straight forward.

(33) a. WS= [R, teksto] where R=basa
b. MERGE (R, teksto, WS) = WS' = [{R, teksto},…]
c. MERGE (v, {R, teksto}, WS) =WS'= [{v, {R, teksto}},…]
c. MERGE (EA, {v, {R, teksto}}, WS) = WS' = [{EA, {v, {R, teksto}}},…]
d. MERGE (T, {EA, {v, {R, teksto}}}, WS) = WS' = [{T, {EA, {v, {R, teksto}}}, …]

The corresponding tree diagram of this derivation is given in (34)

(34) *Tree Diagram Illustrating Illustrating Intransitive Construction Adapted from Aldridge (2006)*



The difference between the derivations in (29) and (33) lies in the position of the IA. In (29), the IA moves to the outer edge of v* making it multiple-specifier construction while in (33), the IA remains in-situ. This difference in the position of IA is accounted for as the checking of Extended Projection Principle or the EPP feature of v* as asserted by Aldridge (2006). EPP, however, has been eliminated in the new formulation of MERGE in favor of the labeling algorithm (LA) (Epstein as cited in DMC KeioUniversity (2023)). In the LA approach, movement or internal merge of SOs to another position is motivated by the necessity of SOs to be labeled. I put aside discussion on how the LA approach can be implemented in the analysis of this movement or internal merge as such analysis lies beyond the scope of the present study and could be a topic for future research.

Wh-question Formation in Tagalog/Filipino

Similar to English, *wh*-question formation in Tagalog/Filipino involves the fronting of the target constituent to the front of the sentence as shown in (34).

(35) a.	Binili	ng	guro	ang	tela
	<IA. Foc> PERFC.TRV-buy	ERG	teacher	ABS	cloth
	The teacher bought the cloth.				

b.	Ano	ang	binili	~~ang tela~~	ng	guro?
	What	FINV	<IA. Foc> PERFC.TV-buy	~~ABS cloth~~	ERG	teacher

c. What did the teacher buy?

In (35), the phrase *ang tela* is the target of the question i.e., the information being elicited. This phrase is replaced by the *wh*-word *Ano* which is used for inanimate NP. This *wh*-word is then moved to the front of the sentence as illustrated in (36).

(36) Ano ang binili ~~ang tela~~ ng guro?

Another operation that may be involved in the *wh*-question formation is the insertion of another *ang* placed after the *wh*-word and before the verb. This *ang* can be considered as a different function word from the commonly classified preposition ANG placed before an argument that indicates focus. This *ang* behaves similarly with the non-relation inversion marker *ay*. *Ay* inverts the order of Tagalog sentence from comment-topic to topic-comment order (Ramos, 1971).

(37) a. Comment-topic:	Pupunta	sya	sa	bahay.
	<EA.foc>PROSP. INTR-Go	3SG	LOC	house

b. Topic-comment:	Sya	ay	pupunta	sa	bahay.
	3SG	NFINV	<EA.foc>PROSP. INTR-Go	LOC	house

(Adopted from Ramos (1971))

I assume that Tagalog/Filipino has two inversion markers, *ay* and *ang*. The former is non-focus while the latter is focus. The evidence that *ang* functions as an inversion marker can be shown in equative sentences.

(38) a. Comment-topic:	Uminom	ang	lalake	ng	alak.
	<EA.foc> PERFC.INTR-drink	ABS	man	ERG	liquor

b. Topic-comment:	Ang	lalake	ang	uminom	ng	alak.
	ABS	man	FINV	<EA.foc> PERFC.INTR-drink	ERG	liquor

Like *ay*, *ang* reverses the order of comment, and topic and gives emphasis on the topic. I gloss the *ang* inversion marker as FINV (Focus Inversion Marker). The addition of the focus inversion marker also applies in *wh*-question in Tagalog/Filipino.

To illustrate how *wh*-question formation in Tagalog/Filipino can analyzed within the current MERGE framework, we repeat the resultant WS of the sentence *Binili ng guro ang tela* in (39)

(39) a. WS= [{T, {IA, {EA, {v*, {R, IA}}}}}]
b. WS= [C, {T, {IA, {EA, {v*, {R, IA}}}}}
c. MERGE (C, {T, {IA, {EA, {v*, {R, IA}}}}}, WS) = WS’= [{C, {T, {IA, {EA, {v*, {R, IA}}}}}}, …]
d. WS= [IA, {C, {T, {IA, {EA, {v*, {R, IA}}}}}}]
e. MERGE (IA, {C, {T, {IA, {EA, {v*, {R, IA}}}}}}, WS] =WS’ = [{IA, {C, {T, {IA, {EA, {v*, {R, IA}}}}}}},…]
f. FC ({*ang, tela*}, {*ang, tela*})
g. WS’ =[{{ang, tela}, {C, {T, {EA, {v*, {R, {~~ang, tela~~}}}}}}}},…]

The derivation begins with the basic verb-initial sentence structure in (39a) in which the IA is already internal merged to the outer edge of v* as per Aldridge (2006). This is followed by the addition of C(omplementizer) to the WS in (39b). This C is the focus inversion marker *ang*. After *ang* is merged in (39c), IA is selected and copied to the WS via minimal search in (39d). It is then merge in (39e). Since the phase CP is complete, the operation form copy or FC applies to assign a copy relation to the two instances of IAs in (39f) which deletes the lower copy in (39g) indicated by a grey area and a strikethrough. *Ang tela* is then externalized as *Ano* completing the derivation.

For intransitive constructions in (40), the *wh*-question formation is given in (41)

(40) a. Sumulat ang mag-aaral ng tula.
<EA. foc> PERFC.INTR-write ABS student ERG poem.
The student wrote a poem.
b. Sino ang sumulat ng tula?
What FINV <EA. foc> PERFC.INTR-write ERG poem?
What did the student write?

(41) a. WS = [T, {EA, {v, {R, IA}}}]
b. MERGE (T, {EA, {v, {R, IA}}}, WS) = WS’ = [{T, {EA, {v, {R, IA}}}},…]
c. WS = [C, {T, {EA, {v, {R, IA}}}}]
e. MERGE (C, {T, {EA, {v, {R, IA}}}}, WS) = WS’ = [{C, {T, {EA, {v, {R, IA}}}}}, …]
f. WS = [EA, {C, {T, {EA, {v, {R, IA}}}}}]
g. MERGE (EA, {C, {T, {EA, {v, {R, IA}}}}}, WS) = WS’ = [{EA, {C, {T, {EA, {v, {R, IA}}}}}}, …]
h. FC ({ang, mag-aaral}, {ang, mag-aaral})
g. WS’ = [{{ang, mag-aaral}, {C, {T, {{~~ang, mag-aaral~~}, {v, {R, IA}}}}}}, …]

In this particular derivation, the EA is selected and copied in the WS via Minimal Search in (41f). It is then internal merged in (41g). FC then assigns copy relation the two inscriptions of {ang, mag-aaral} and deletes the lower copy while higher copy is externalized as *sino*. In both derivations in (39) and (41), the internal merge of the target argument to the specifier of C follows the 3rd factor principle of Minimal Search i.e., the most prominent term is the one subject to internal merge, the IA in (39) and the EA in (41).

Equative Sentences

Equative sentences are sentences in which the ANG-marked NP occurs at the beginning of the sentence.

(42) a. Ang guro ang bumili ng tela.
ABS teacher <EA. foc> PERFC.INS-buy ERG cloth
It was the teacher who bought the cloth.
The teacher was the one who bought the cloth.

According to Ceña and Nolasco (2011), there are two approaches to describing equative sentences in Tagalog/Filipino, the "Merge" and the "Movement" approach—terms that are obsolete in the recent minimalist literature. I recast these approaches using the terms of the current formulation of MERGE. I also argue that MERGE, the concept of inscription, FC and LSC provide insight into which of the two approaches is correct in relation to the SMT. The approximate illustrations of the "Merge" and the "Movement" approaches are given in (43).

(43) a. Numeration= [ang, bili, -um, guro, tela]

NP

{{Ang, guro}$_i$, {ang, {bumili, {PRO$_i$, {ng, tela}}}}}

b. {{Ang, guro}, {ang, ~~{ang, guro}~~, {bumili, {ng, tela}}}}

In (43a), the numeration is accessed to assemble the NP, *ang guro*. This NP is then merged to the SO already constructed. This NP binds control pronoun internal to the clause to capture the locality condition required in the theta role assignment. On the other hand, in (43b), the NP, {*ang*, *guro*} is accessed via Minimal Search in the previously constructed SO, copied and merged to the same SO. The derivation in (43a) has a resemblance to the inscription-based analysis of control constructions while (43b) is a simple application of internal merge.

The inscription-based analysis of control construction is an approach proposed by Chomsky et al. (2023) to eliminate control pronouns. Inscriptions are a mere representation of the original lexical items that remain in the lexicon. In other words, the WS can only contain inscription, not the actual lexical items. Thus, multiple inscriptions of the same lexical item can be used during the course of the derivation. In their control analysis, Chomsky et al. (2023) hypothesized that a sentence with subject control is a result of the EM of another inscription.

(44) a. WS= [{to, {EA, sleep}}, …] where EA= {the, man}
b. WS= [{try, {to, {EA, sleep}}}, …]
c. WS= [{v*, {try, {to, {EA, sleep}}}}, …]
d. WS= [{EA, {v*, {try, {to, {EA, sleep}}}}}, …]

In this derivation, the EA is first merged with the verb, *sleep* and then merged with infinitival *to* in (44a). The matrix verb *try* and the verbalizer v* are then merged into the rest of the structure in (44b) and (44c). The EA in the lower clause can then move to the edge of v* to satisfy the theta role assignment of the root *try*. This movement, however, is illicit under the new formulation of MERGE because of the duality of semantics.

(45) Duality of Semantics (*adopted from Chomsky et al., 2023, p. 33*)
EM, and only EM creates theta position.

Since the edge of v* is a theta position, it can only be filled by an inscription via EM. This can only be accomplished by accessing the lexicon and assembling another inscription of the EA. This EA is then merged to the rest of the structure in (44d). FC then applies and assigns copy relation to the identical inscriptions which deletes the lower copy at the sensorimotor interface. PIC also ensures that the MY is followed in this the derivation.

(46) a. FC (EA, EA)

b. WS' = [{EA, {v*, {try, {to, {~~EA,~~ sleep}}}}}, …]

Recasting (43a) using the inscription-based analysis of control, the derivation and the respective WSs for the EM approach to equative sentence in (42) are illustrated below:

(47) a. WS= [{bumili, {{ang, guro}, {ng, tela}}}]

b. WS= [{{ang, guro}, {bumili, {{ang, guro}, {ng, tela}}}} where {ang, guro} is a new inscription taken from the lexicon.

c. FC ({ang, guro}, {ang, guro})

d. WS= [{{ang, guro}, {bumili, {~~{ang, guro}~~, {ng, tela}}}}

This EM approach to Equative Sentences roughly corresponds to the "Merge" approach in Ceña and Nolasco (2011). The "Movement" approach to this syntactic phenomenon, on the other hand, can be captured by a simple application of internal merge (henceforth IM approach). The derivation in (47) illustrates the "Movement" approach recast as an application of IM to the WS.

(48) a. WS= [{bumili, {{ang, guro}, {ng, tela}}}]

b. Search ({ang, guro}, TP} where TP= {bumili, {{ang, guro}, {ng, tela}}}

c. MERGE ({ang, guro}, TP, WS)= WS' = [{{ang, guro}, TP}]

d. FC ({ang, guro}, {ang, guro})

e. WS= [{{ang, guro}, {bumili, {~~{ang, guro}~~, {ng, tela}}}}]

The difference between (47) and (48) lies in how the copy of {ang, guro} is accessed and merged with the rest of the structure. In (47), this is accomplished by accessing the lexicon, assembling an inscription of {ang, guro}, adding it to the WS, and finally external merging it to the previously constructed SO. In contrast, (48) illustrates access to the WS via Minimal Search. When Minimal Search hits the inscription {ang, guro}, it copies it and internal merges it to the previously constructed SO. FC then assigns a copy relation to the identical inscriptions and gives instructions to SM that the lower copy must be unpronounced.

With the EM and IM approaches to equative sentences having been reformulated in the new framework, principled choice about which of the two is correct can be provided. Ceña and Nolasco (2011) favored the movement analysis because of the complication introduced by the control pronoun PRO. Their argument is supported by inclusiveness condition.

(49) Inclusiveness Condition (Chomsky, 1995, p. 209)

Any structure formed by the computation is constituted by elements already present in the lexical items selected for N(umeration).

Chomsky (2008) explained that the inclusiveness condition aims at, "dispensing with bar levels, traces, indices, and similar descriptive technology introduced in the course of derivation of an expression" (p. 138). The inclusiveness condition, thus, renders the control analysis of equative sentences illicit. Furthermore, under the new framework, the EM approach violates the duality of semantics; the edge of TP to which the inscription {ang, guro} is external merged is not a theta position thus such EM is a violation of LSC which leaves the IM approach the viable option for the analysis of equative sentences under the new formulation of MERGE and SMT.

Conclusions

In this paper, I propose derivations of Tagalog/Filipino sentence structures using the new formulation of MERGE, the 3rd factor constraint, and Language Specific Conditions. Drawing from the work of Aldridge (2006) and Ceña and Nolasco (2011), I applied these concepts in the analysis intransitive, transitive, and *wh*-questions. Additionally, I argue that Ceña and Nolasco's (2011) movement analysis of equative sentences is correct as the alternative "Merge" approach to this syntactic phenomenon is illicit based on the new MERGE and the Duality of Semantics i.e., only external MERGE creates a theta position. This shows that current formulation of MERGE provides a principled choice between two competing syntactic analyses of the same phenomenon. The study contributes to the goal of using less commonly studied languages in illustrating, sharpening, and refining theoretical concepts through their application in the analyses of these languages.

Recommendations

The analysis presented in this paper is a preliminary sketch. Further examination into the application of MERGE and SMT on data from less known languages like Tagalog/Filipino is necessary to further sharpen these analytical devices, and demarcate their limitations if any.

Declaration of Competing Interest

No personal or financial conflict of interest has motivated nor arisen from this study.

Declaration of Use of Generative AI / AI-assisted Technologies

The use of AI, particularly ChatGPT, in the completion of this study, was limited to review, clarification of concepts, and refining and polishing of the language. No section, data, or analysis were directly taken from AI.

References

Aldridge, E. 2006. *Absolutive case in Tagalog*. Chicago Linguistic Society 42. Chicago University.

Alexiadou, A., & Lohndal, T. (2021). From the origins of government and binding to the current state of minimalism. In *A Companion to Chomsky*, eds. N. Allott, T. Lohndal, G. Rey, 25-51. Wiley Blackwell.

Bencheikh, A. (2019, June 07). *Noam Chomsky, Fundamental Issues in Linguistics (April 2019 at MIT) - Lecture 2* [Video]. YouTube. https://www.youtube.com/watch?v=GPHew_smDjY

Berwick, B.C. & Chomsky, N. (2016). *Why only us: Language and evolution*. Cambridge, MA: MIT Press.

Ceña, R. M. & Nolasco, R.M.D. (2011). *Gramatikang Filipino:Balangkasan*. Quezon City, Philippines: University of the Philippines Press.

Citko, B. (2000). *Parallel merge and the syntax of free relatives* [Doctoral dissertation]. Stony Brook University.

Citko, B. (2005). On the nature of merge: External merge, internal merge, and parallel merge. *Linguistic Inquiry, 36*, 475–496.

Citko, B. (2014). *Phase theory: An introduction*. United Kingdom: Cambridge University Press.

Citko, B., & Yuksek, M. G. (2020). *Merge: Binarity in (Multidominant) Syntax.* MIT Press.

Chametzky, R.A. (1996). *A theory of phrase markers and extended base*. State University of New York Press

Chomsky, N. (1957). *Syntactic structure*. The Hague, The Netherlands: Mouton.

Chomsky, N. (1995). *The minimalist program*. Cambridge, MA: MIT Press.

Chomsky, N. (2000). Minimalist inquiries: The framework. In *Step by step: Minimalist essays in honor of Howard Lasnik*, eds. R. Martin, D. Michaels, & J. Uriagereka, 89-155. MIT Press.

Chomsky, N. (2001). Derivation by phase. In *Ken Hale: A life in Language,* ed. M. Kenstowicz,1-52. Cambridge, MA: MIT Press.

Chomsky, N. (2004). Beyond explanatory adequacy. In *Structures and beyond: The cartography of syntactic structures*. A. Belleti, 104-131. New York: Oxford University Press.

Chomsky, N. (2005). Three factors in language design. *Linguistic inquiry, 36*(1), 1-22. Doi:10.1162/0024389052993655

Chomsky, N. (2007). Approaching UG from below. In *Interfaces + recursion= Language*, eds. U. Sauerland, & H. Gärtner, 2-29. Mouton De Gruyter.

Chomsky, N. (2008). On phases. In *Foundational Issues in Linguistic Theory*, eds. R. Freidin, C.P. Otero, M.L. Zubizarreta, 133-166. Cambridge, MA: MIT Press.

Chomsky, N., Gallego, A., & Ott, D. (2017). Generative grammar and the faculty of language: Insights, questions, and challenges. *Catalan Journal of Linguistics* (Special Issue), 229–261.

Chomsky, Noam. (2019). The reading lecture. *Catalan Journal of Linguistics (Special Issue)*, 263–285.

Chomsky, N., Seely, T. D., Berwick, R., Fong, S., Huybregts, M.A.C., McInnery, A. & Sugimoto, Y. (2023). *Merge and the strong minimalist thesis*. Cambridge University Press.

DMC KeioUniversity (2023, April 05). *［Lecture 1］ Working toward the Strong Interpretation of SMT* [Video]. YouTube. https://www.youtube.com/watch?v=vD1Vqv7cmq4&t=369s

Epstein, S. D., Kitahara, H., & Seely, T. D. (2022). *A minimalist theory of simplest merge*. New York: Routledge.
Fong, S. (2021). *Some third factor limits on Merge* [Manuscript]. University of Arizona.
Lebeaux, D. (1988). *Language Acquisition and the Form of Grammar* [Doctoral dissertation]. University of Massachusetts.
Nunes, J. (2001). Sideward movement. *Linguistics Inquiry*, *32*, 303–344.
Pinker, S. (1984). *The Language instinct*. Penguin Books
Ramos, T. (1971). *Tagalog structures*. University of Hawaii Press.